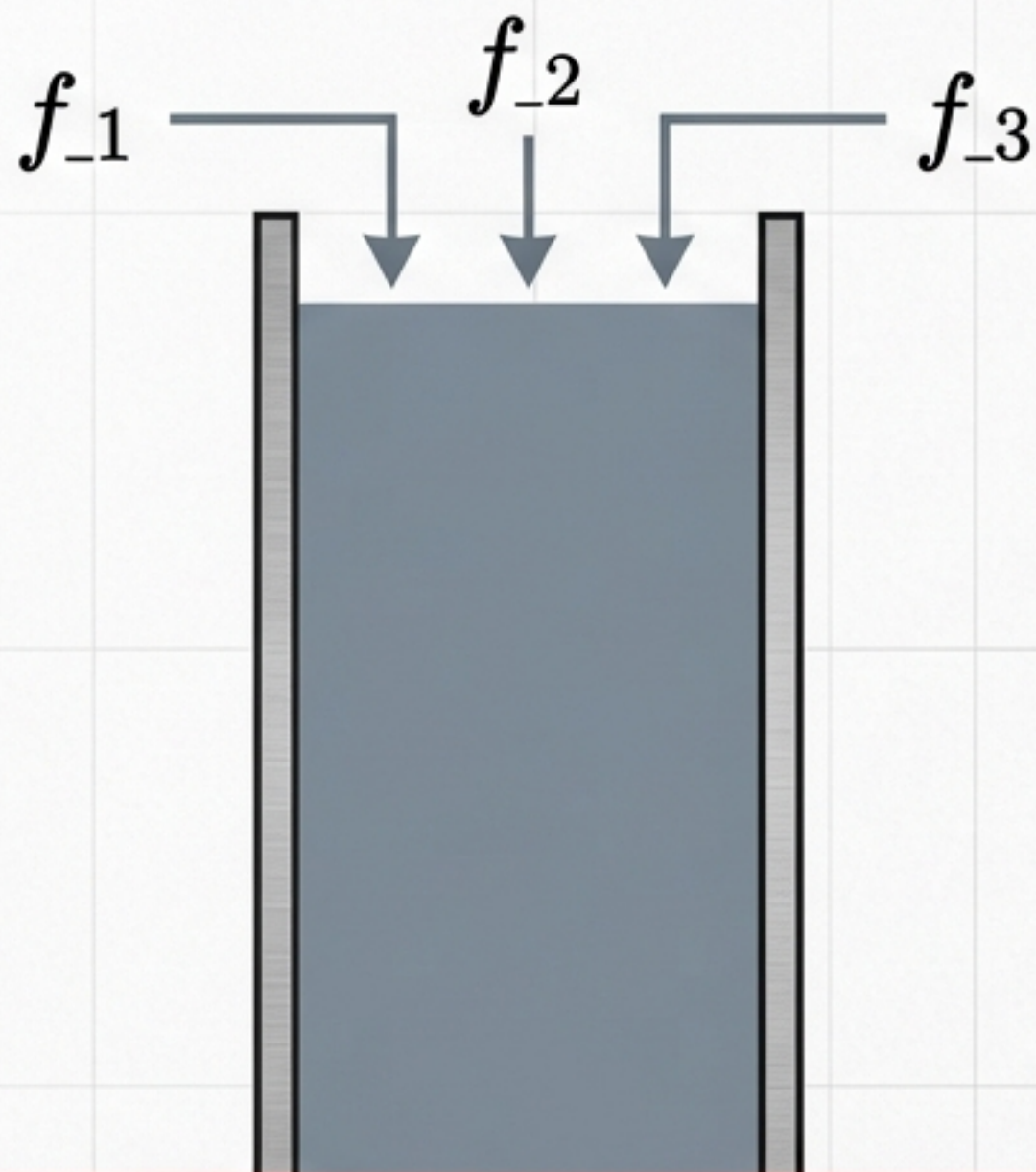


Attention Residuals

Dynamic Depth-wise Routing
for Large Language Models

A STRUCTURAL BLUEPRINT OF MODERN LLM ARCHITECTURE REIMAGINED.

The Hidden Bottleneck: Unweighted Depth Accumulation



PreNorm Dilution: Output magnitudes grow as $O(L)$, burying early-layer signals.

Fixed Accumulation

Every layer receives the exact same uniformly-weighted sum of prior outputs.

No Selective Retrieval

A deeper layer cannot selectively suppress noise or isolate specific early representations.

Unbounded Growth

Forces deeper layers to learn increasingly massive outputs to influence the accumulated stream.

The Duality of Time and Depth

Sequence (Time) Axis

RNNs

Fixed, compressed state over time.



Transformers

Softmax Attention over all previous tokens.



Depth Axis

ResNets

Fixed, accumulated state over depth.

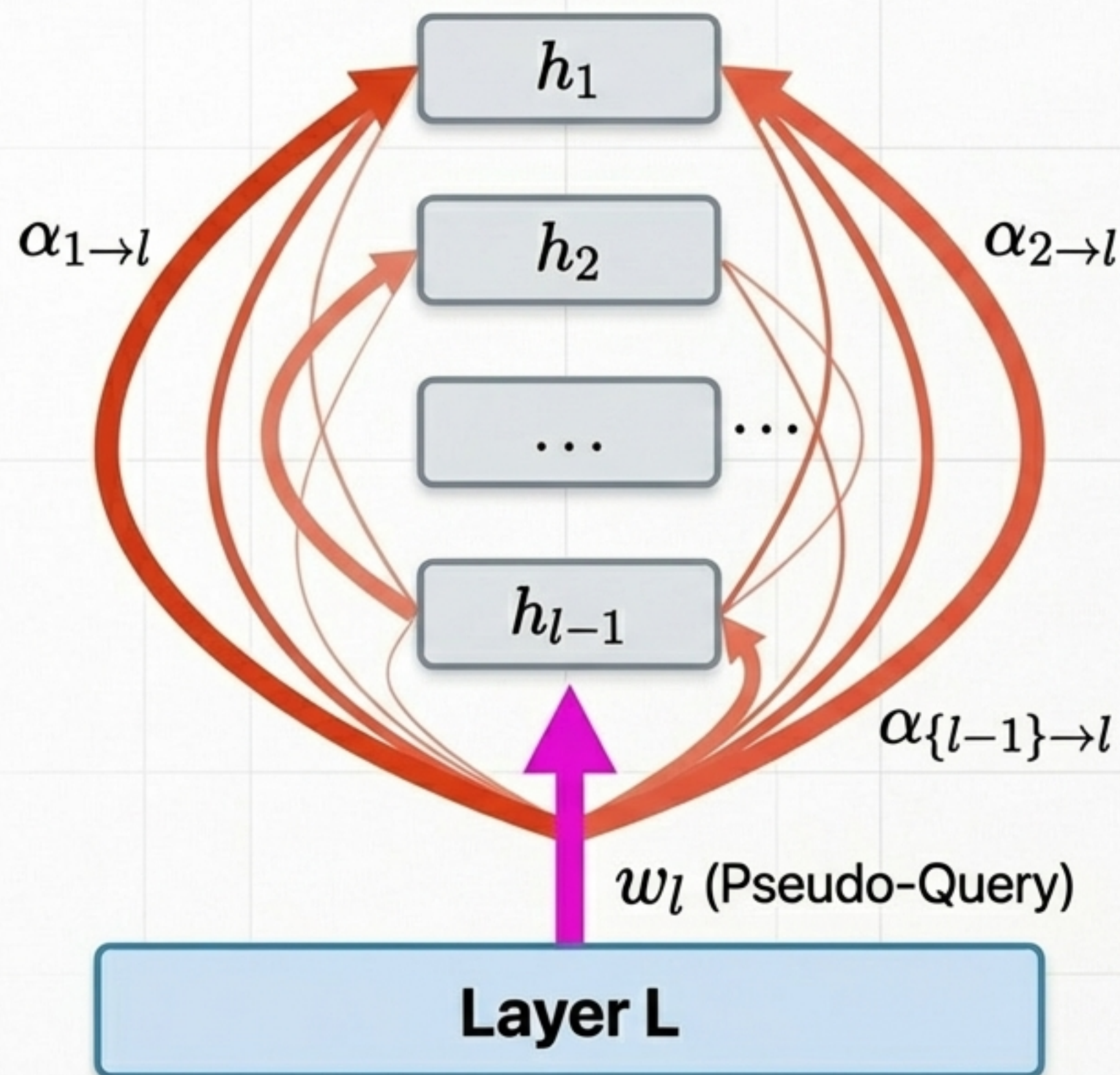


Attention Residuals

Softmax Attention over all previous layers.

Central Synthesis: We revolutionized sequence modeling by replacing fixed recurrence with dynamic routing. Depth-wise aggregation remains trapped in a recurrent bottleneck. **AttnRes** completes the linear-to-softmax transition for network depth.

The Theoretical Ideal: Full Attention Residuals



The Mechanism

Fixed accumulation is replaced by

$$h_l = \sum \alpha_{i \rightarrow l} \cdot v_i$$

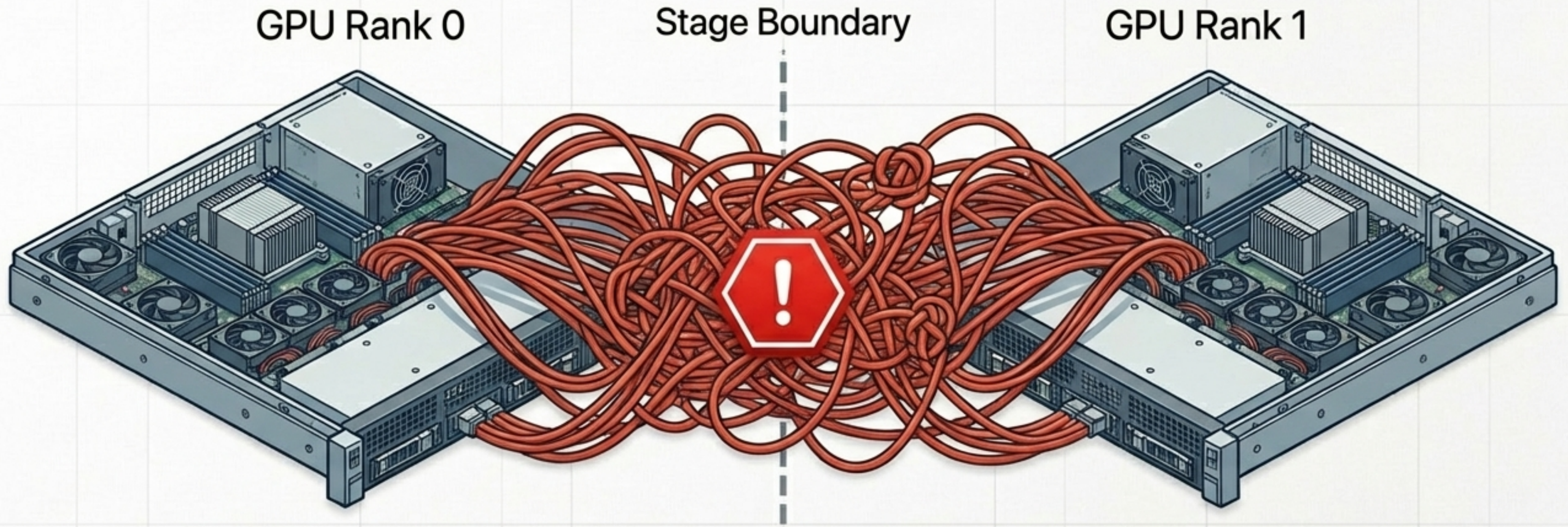
Decoupled Queries

The pseudo-query w_l is a single learned d -dimensional vector per layer, computed independently of the layer's forward pass.

RMSNORM Stabilization

Prevents layers with massive natural outputs from monopolizing attention probabilities during routing.

The Scaling Wall: Memory and Pipeline Communication



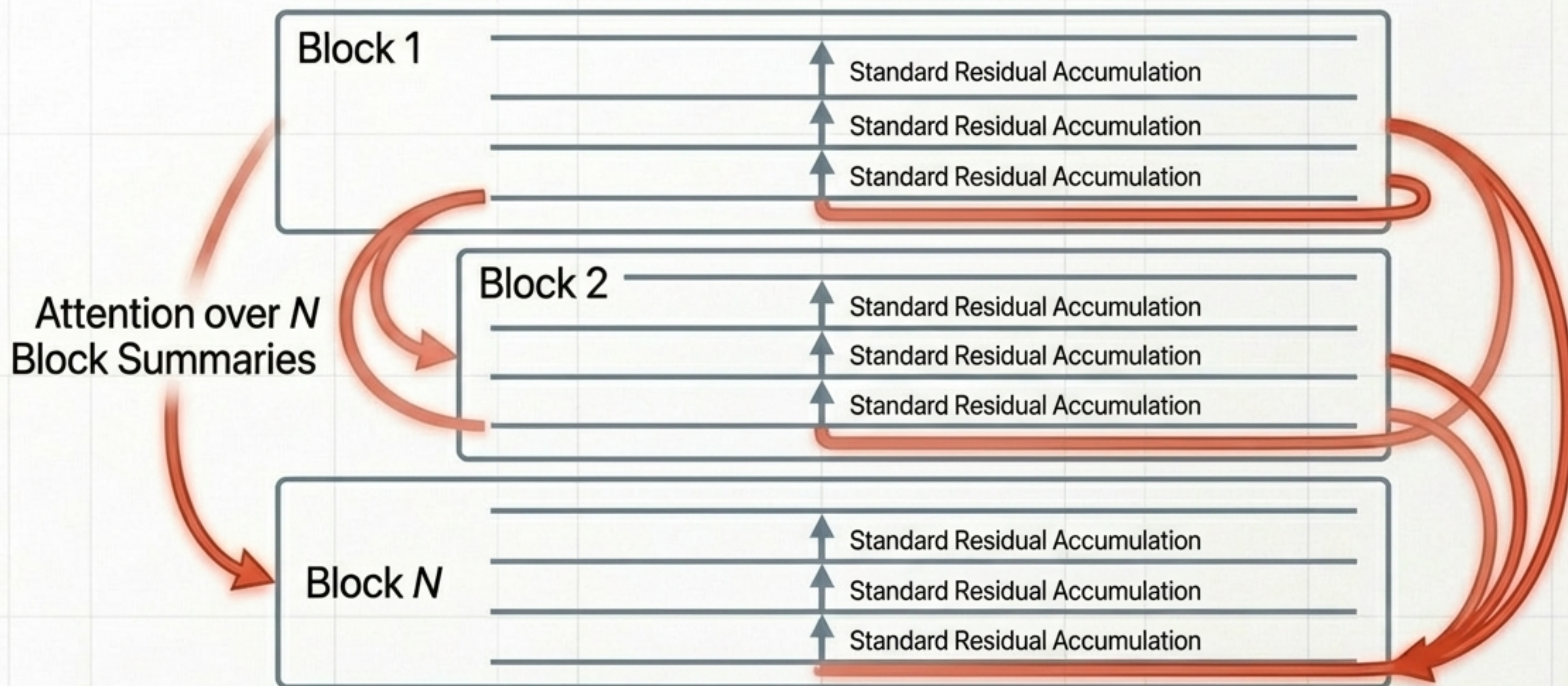
Pipeline Bottleneck

Maintaining individual layer outputs inflates cross-stage communication overhead to $O(L_d)$ per token.

Memory Obliteration

Under activation recomputation regimes, preserving L individual states destroys all memory efficiency gains.

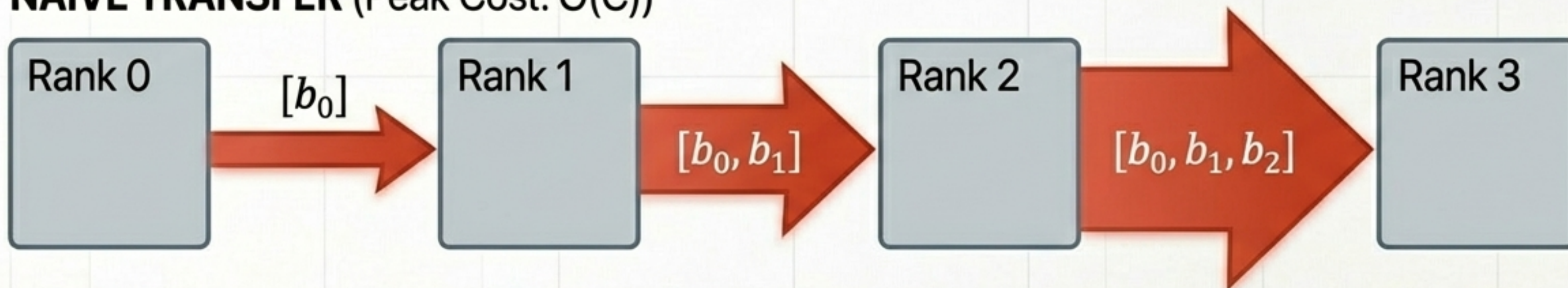
The Practical Reality: Block Attention Residuals



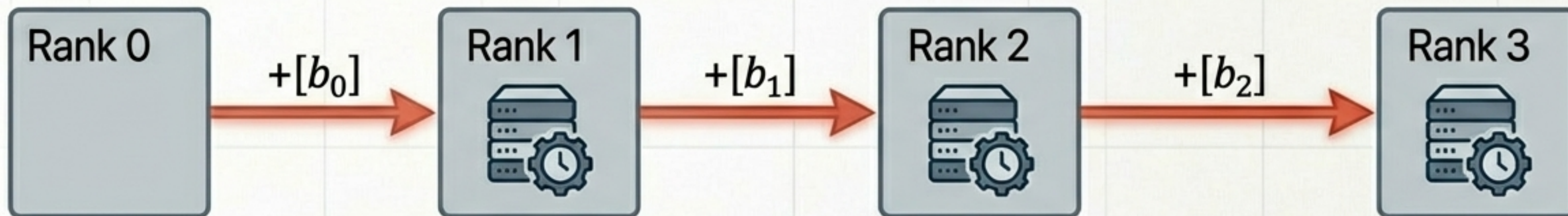
System Impact: Memory and communication costs plummet from $O(L_d)$ to $O(N_d)$. For a 128-layer model, storing 8 block summaries perfectly balances expressivity with strict hardware constraints.

Infrastructure Magic I: Cross-Stage Caching

NAIVE TRANSFER (Peak Cost: $O(C)$)



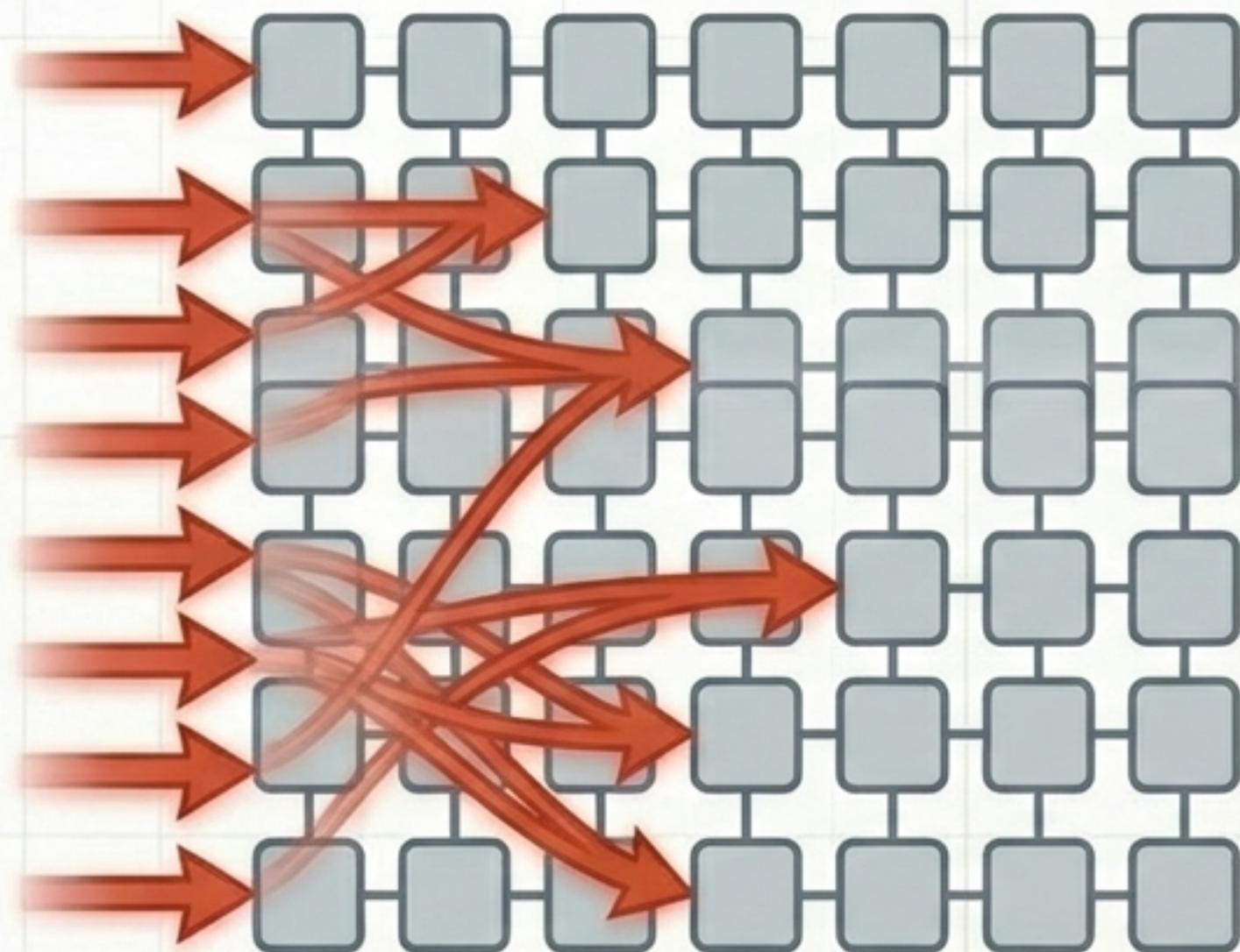
CROSS-STAGE CACHING (Peak Cost: $O(P)$)



The Result: Caching eliminates redundant block transmissions. End-to-end training overhead drops to $<4\%$.

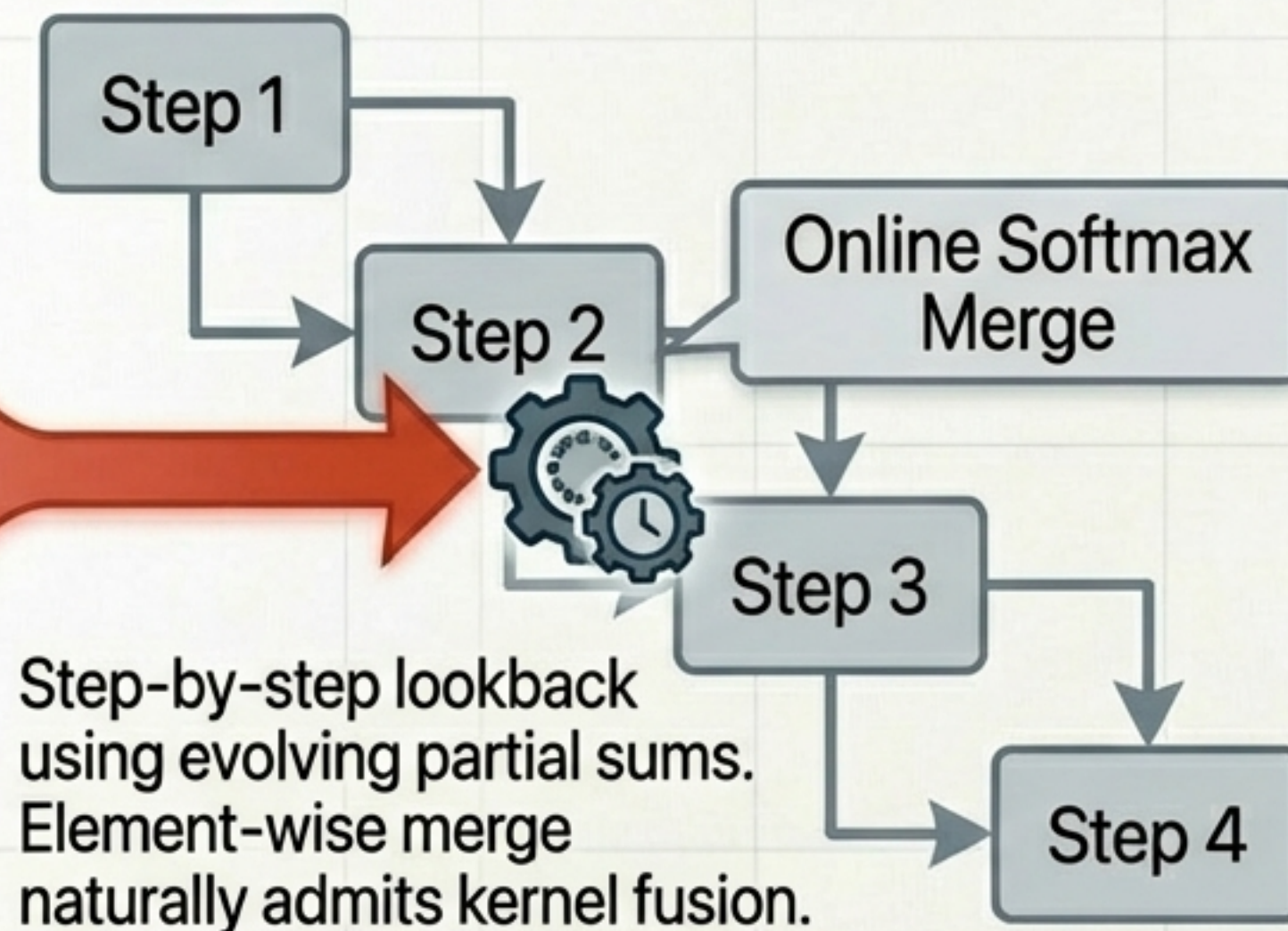
Infrastructure Magic II: Two-Phase Inference

Phase 1: Parallel Inter-Block Attention



Batched queries fire simultaneously against cached block representations. Amortizes heavy HBM memory reads to just ONCE per block.

Phase 2: Sequential Intra-Block Attention



<2% Latency Overhead

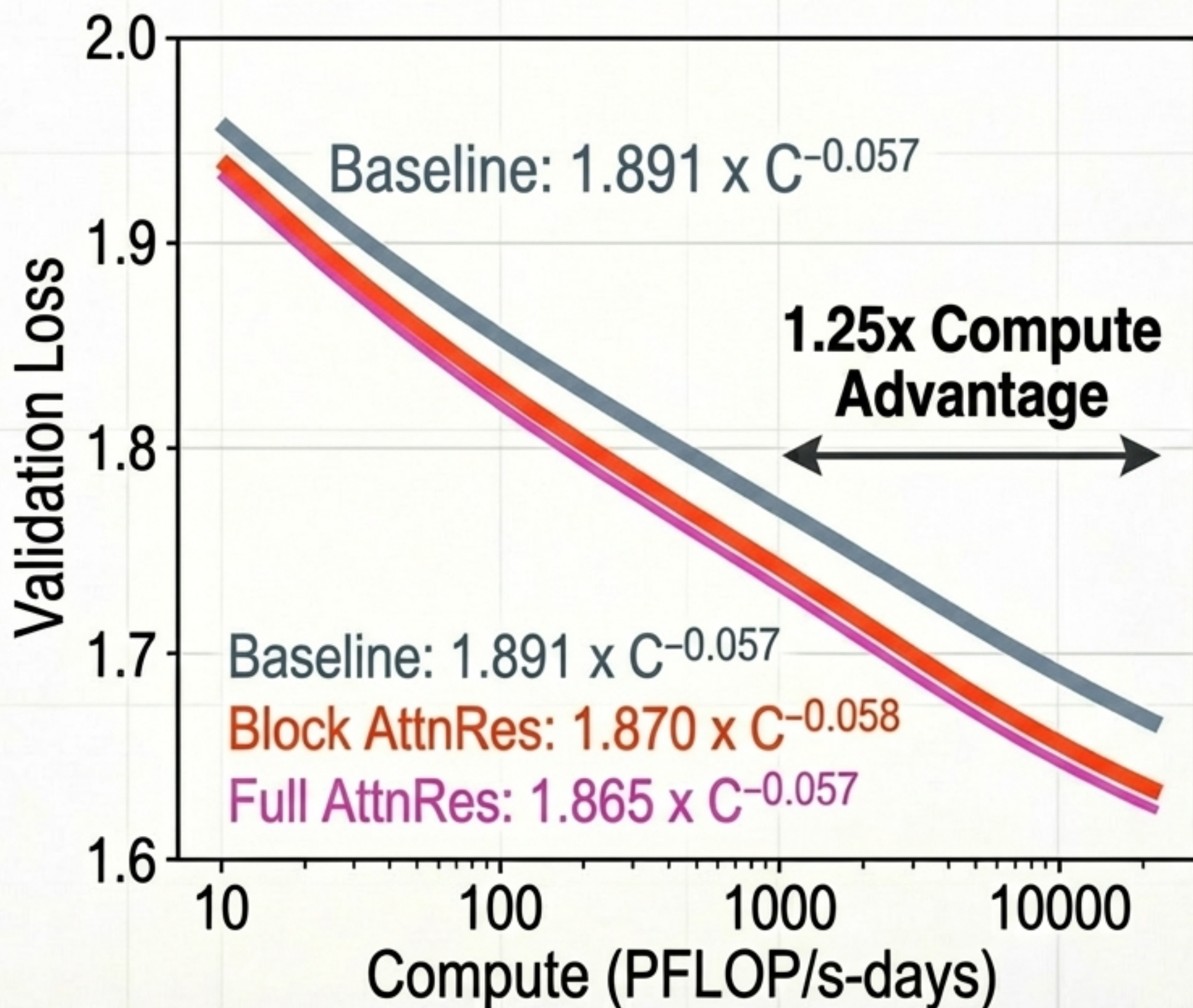
End-to-end inference latency is tightly contained, keeping autoregressive decoding blazingly fast.

The Residual Landscape: A Diagnostic Matrix

Architecture	Weight Type	Source Access	Memory I/O per Layer	Hardware Scalability
Standard Residual	Fixed	Immediate Predecessor	3d	High
Highway / DeepNorm	Dynamic	Immediate Predecessor	3d	High
Multi-stream (mHC)	Dynamic	Immediate Predecessor	34d	Low
Full AttnRes	Dynamic	All Previous Layers	24d	Low
Block AttnRes	Dynamic	All Previous Blocks	5.5d	High

Prior methods like mHC achieved dynamic routing via heavy multi-stream recurrences (high I/O). Block AttnRes achieves true cross-layer access via dynamic softmax, keeping memory I/O remarkably lean at 5.5d.

Empirical Proof I: Consistent Scaling Laws



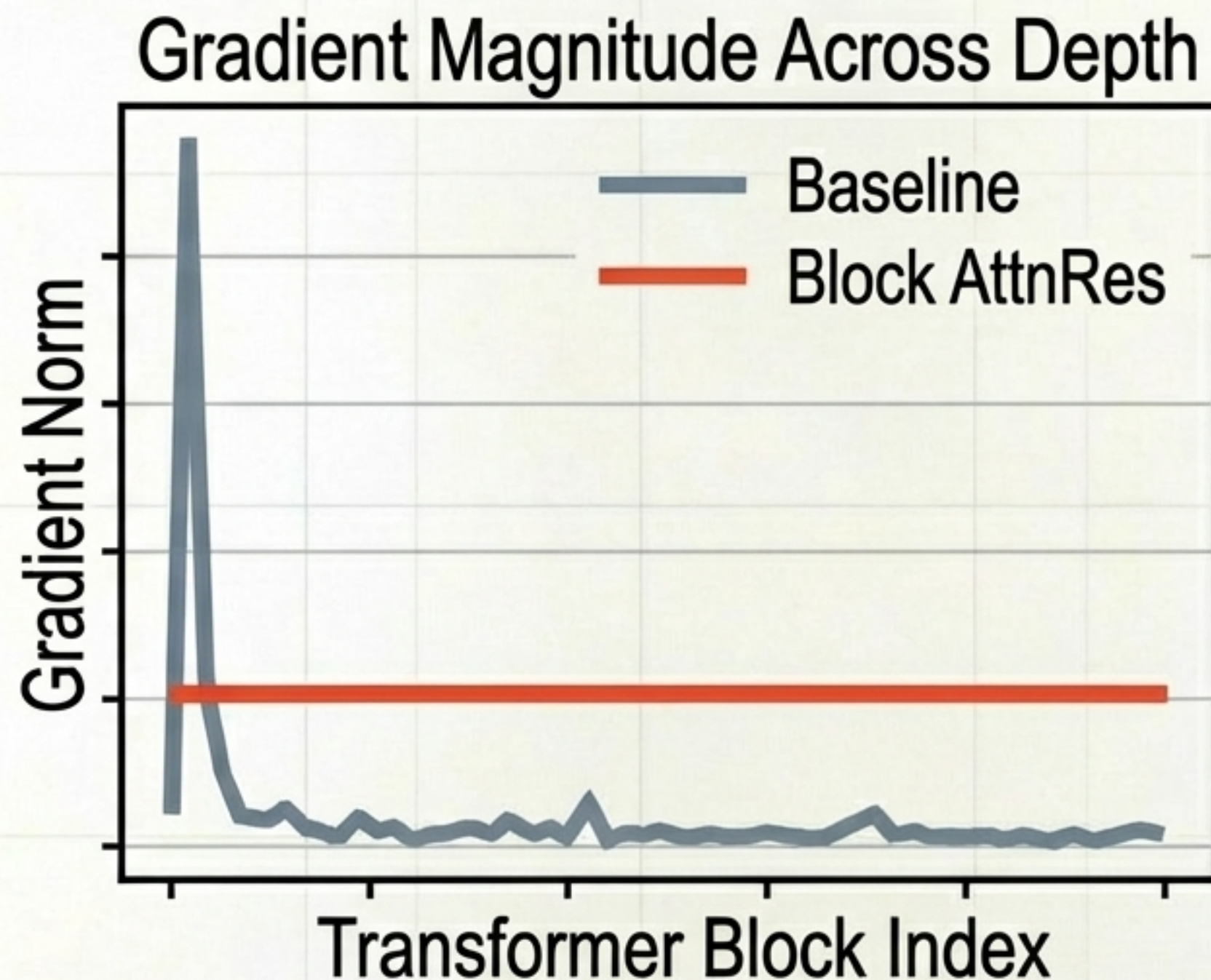
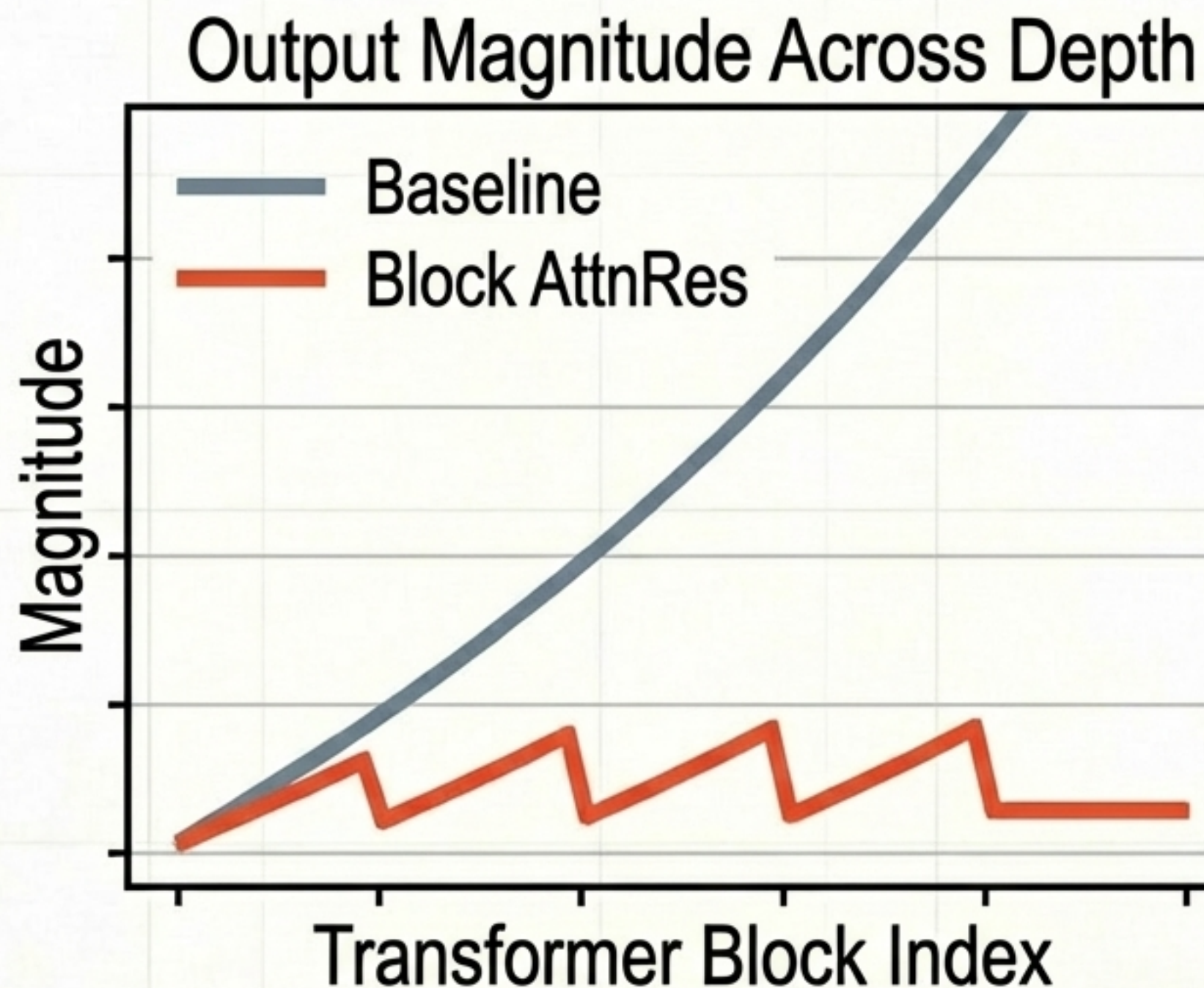
Universal Improvement

Block AttnRes consistently beats the PreNorm baseline across all five evaluated model scales, effectively yielding a 25% compute advantage.

Approaching the Ideal

The gap between Block and Full AttnRes narrows with scale, proving the N-block grouping captures nearly all dynamic routing benefits.

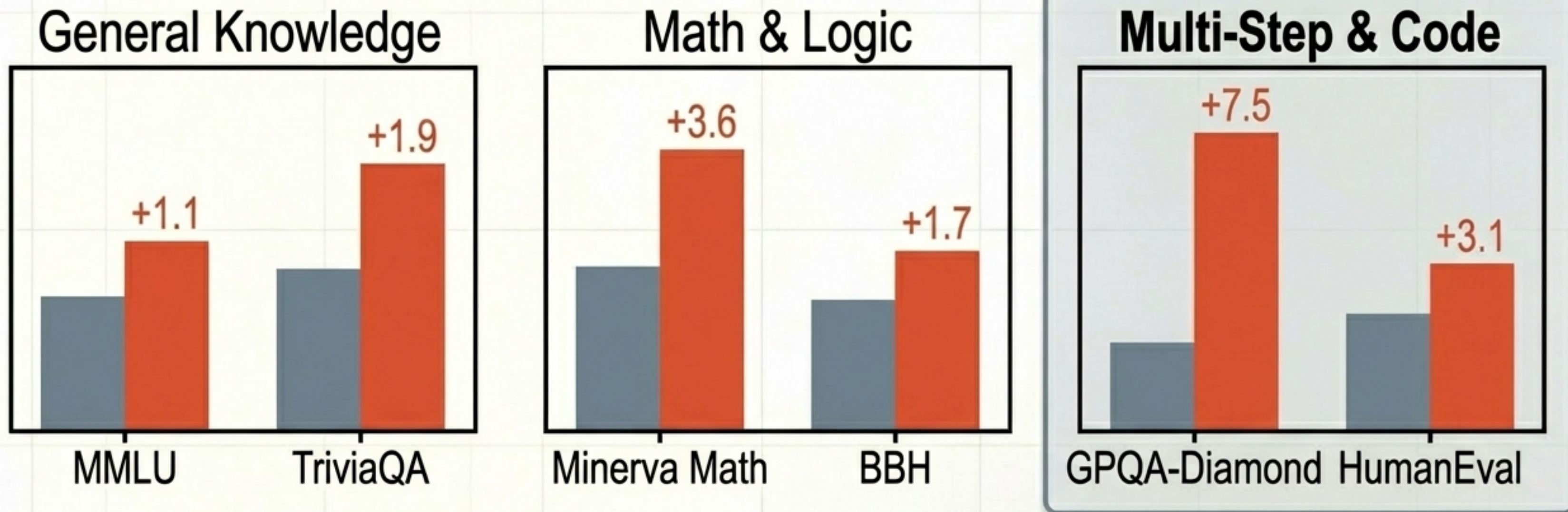
Empirical Proof II: Curing PreNorm Dilution



The Diagnosis **Confirmed**: Dynamic softmax weights introduce competition for probability mass, automatically bounding output growth and forcing uniform gradient distribution.

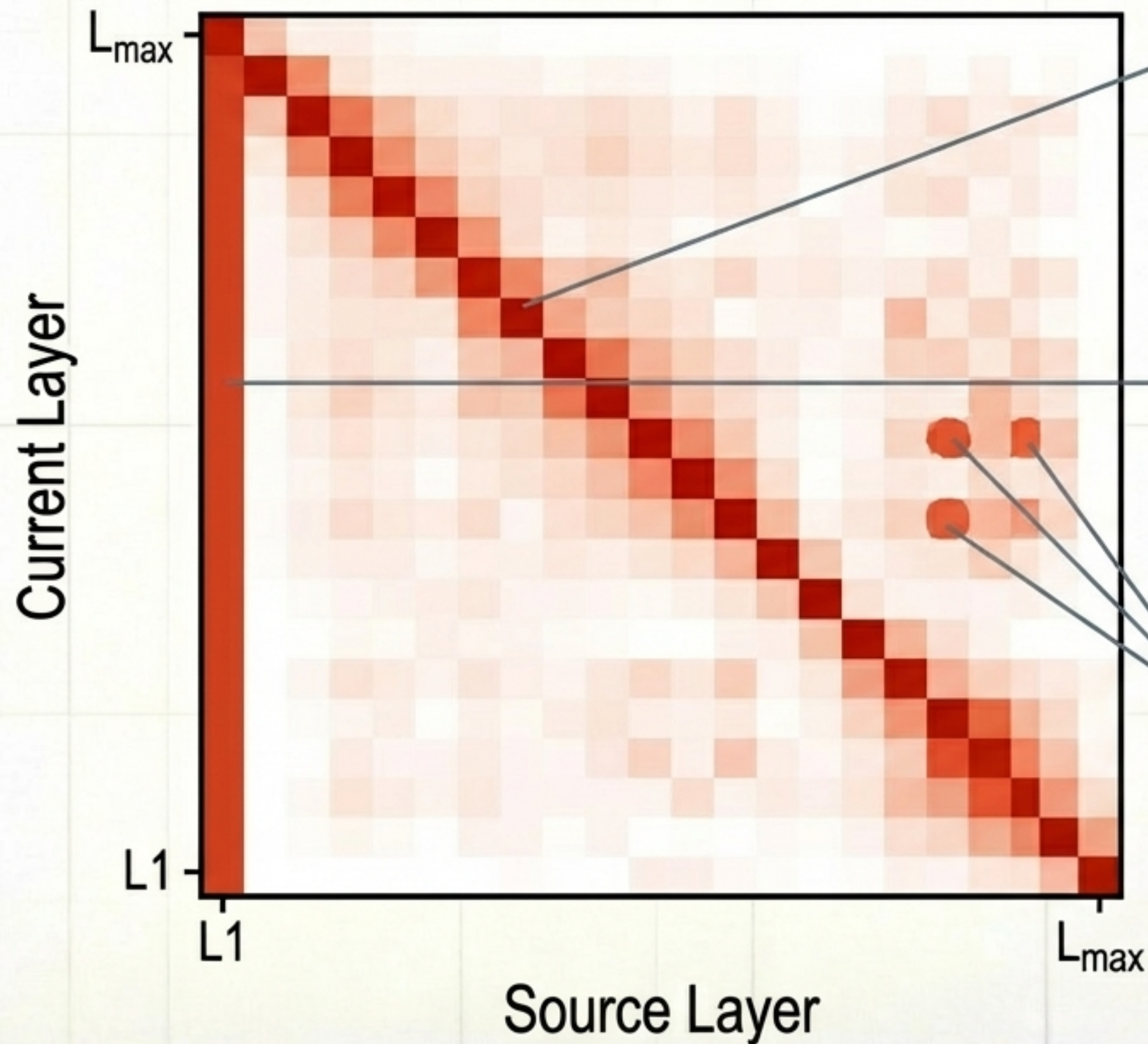
Empirical Proof II: Downstream Reasoning at 48B Scale

Kimi Linear 48B Architecture, pre-trained on 1.4 Trillion tokens.



Synthesis Insight: The ability to selectively retrieve exact, unmodified representations from distant early layers yields outsized gains in complex, multi-step compositional reasoning.

Inside the Black Box: Learned Routing Behaviors



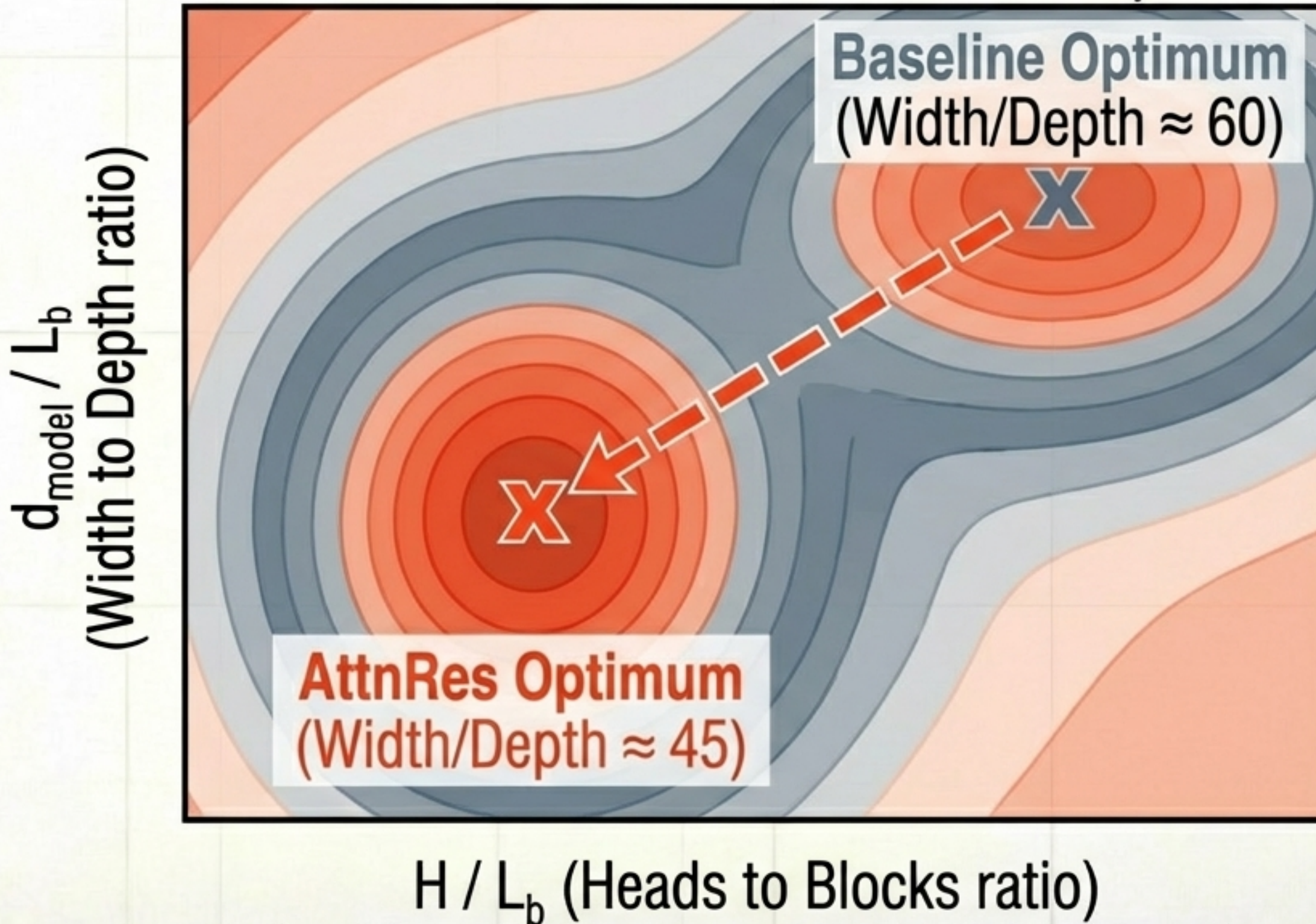
1. **Preserved Locality:** Layers predominantly build on immediate predecessors, preserving standard sequential processing.

2. **Persistent Embedding:** The raw token embedding (h_1) acts as an attention sink, frequently consulted by deep pre-attention layers.

3. **Emergent Skips:** Models learn discrete, long-range skip connections to retrieve early features and bypass intermediate noise.

Redrawing the Optimal Architecture

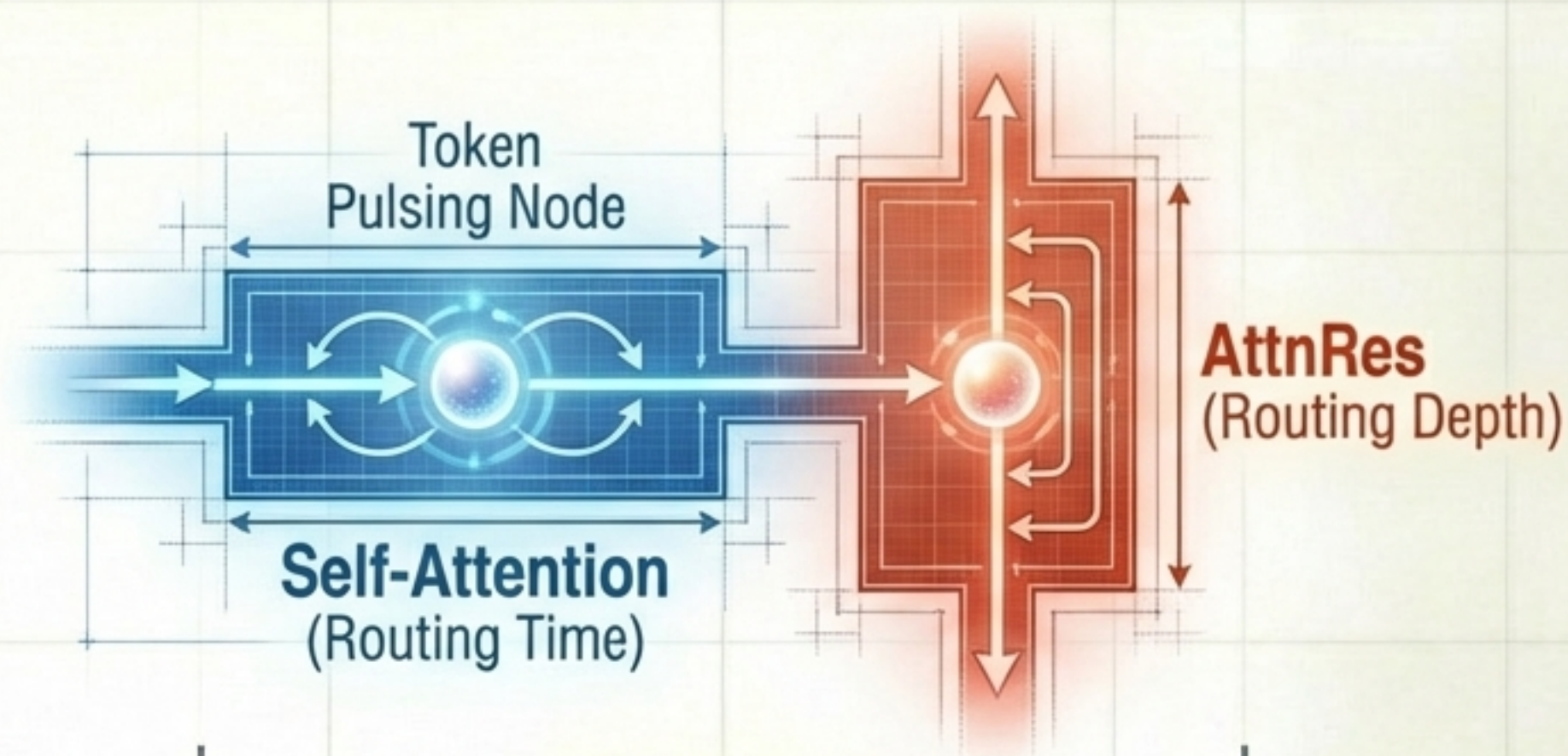
Validation Loss under Fixed Compute



The Design Implication

Because Attention Residuals manage depth stability so effectively—without gradient vanishing or magnitude explosion—the architecture inherently favors models that are significantly deeper and narrower than traditional transformer heuristics advise.

The Future of the Transformer is Smart on Both Axes



The Diagnosis

Fixed residual accumulation acts as a rigid pipe, diluting early signals and destabilizing network depth.

The Cure

Softmax depth attention dynamically routes information, scaling practically via $O(N_d)$ Block designs and 2-phase inference.

The Proof

Delivers a strict 1.25x compute advantage and massive downstream gains (+7.5 GPQA) at 48B scale.

Information in LLMs is no longer forced through a rigid pipe; it is dynamically retrieved exactly when and where it is needed.